

Measures of Centrality and Spread

Your Course or Institution

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Why Summarize Data?

- Raw datasets can be large and complex.
- Summaries help us quickly understand the main tendencies.
- Two key aspects:
 - Centrality: where data tends to be concentrated
 - Spread: how much the data varies
- Focus today: continuous variables (Height, Score) and categorical (Gender, Group)

Centrality vs Spread

- Centrality measures: mean, median, mode
- Spread measures: range, IQR, variance, standard deviation
- Together they describe the “typical” value and variability of a variable
- Visual interpretation complements numeric measures

Mean (Average)

- Sum of all values divided by the number of observations
- Simple formula: $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$
- Interpretation:
 - Represents the "balance point" of data
 - Sensitive to very high or low values (outliers)
- Property of linearity:
 - Mean of $ax + b$ is $a\bar{x} + b$

Mean in R

```
data <- read.csv("data.csv")  
  
# Compute mean for continuous variables  
mean(data$Height)  
mean(data$Score)
```

Median

- Middle value when data are sorted
- Robust to outliers
- For even number of observations: average of two middle values
- Useful to compare with mean:
 - If mean $>$ median $\rightarrow$ right skewed
 - If mean $<$ median $\rightarrow$ left skewed

Median in R

```
median(data$Height)  
median(data$Score)
```


Mode

- Value that occurs most frequently
- Important for categorical variables
- Can be more than one mode (bimodal)

Mode in R

```
getmode <- function(v) {  
  uniqv <- unique(v)  
  uniqv[which.max(tabulate(match(v, uniqv)))]  
}  
  
getmode(data$Gender)
```

Comparing Mean, Median, and Mode

- Mean: sensitive to extremes
- Median: robust, better for skewed distributions
- Mode: shows most frequent category
- Practical tip: report mean and SD for symmetric, median/IQR for skewed

Range and Interquartile Range (IQR)

- Range = max - min; simple but sensitive to outliers
- IQR = $Q3 - Q1$; measures middle 50% of data
- Robust measure, ideal for skewed distributions

Range and IQR in R

```
range(data$Height)  
IQR(data$Height)
```

Variance and Standard Deviation

- Variance: average squared deviation from mean
- Standard deviation: square root of variance, same units as data
- Sensitive to outliers
- Interpretation:
 - SD small $\rightarrow$ data close to mean
 - SD large $\rightarrow$ data spread out

Variance and SD in R

```
var(data$Height)  
sd(data$Height)
```

Effect of Linear Transformations

- If $y = ax + b$, then:
 - $\text{mean}(y) = a \cdot \text{mean}(x) + b$
 - $\text{sd}(y) = |a| \cdot \text{sd}(x)$
- Example: converting cm $\rightarrow$ m

Boxplot - Centrality and Spread

```
library(ggplot2)
ggplot(data, aes(y = Height)) +
  geom_boxplot(fill = "lightblue") +
  labs(title = "Boxplot of Height")
```

Histogram with Mean and Median

```
ggplot(data, aes(x = Height)) +  
  geom_histogram(binwidth = 5, fill = "lightgreen", color =  
    "black") +  
  geom_vline(aes(xintercept = mean(Height)), color = "red",  
    linetype="dashed") +  
  geom_vline(aes(xintercept = median(Height)), color = "blue",  
    linetype="dotted") +  
  labs(title = "Histogram of Height with Mean (red) and  
    Median (blue)")
```

Interpreting Boxplot and Histogram

- Boxplot shows median (line), IQR (box), whiskers, and outliers
- Histogram shows overall shape, skewness, and approximate central tendency
- Comparing mean and median helps understand skewness

Exercise 1

- Compute mean, median, mode, SD, and IQR for:
 - Height
 - Score
- Compare mean vs median
- Identify any outliers

Exercise 1 - Solution

```
# Load data
data <- read.csv("data.csv")

# Compute centrality measures
mean(data$Height)
median(data$Height)

# Compute spread measures
sd(data$Height)
IQR(data$Height)

# Mode function
getmode <- function(v) {
  uniqv <- unique(v)
  uniqv[which.max(tabulate(match(v, uniqv)))]
}
getmode(data$Gender)
```

Interpretation:

- Mean Height: X cm → average “center” of distribution
- Median Height: Y cm → robust measure of centrality
- SD Height: Z cm → how dispersed Heights are
- IQR Height: W cm → middle 50% of data
- Mode Gender: "Male" → most frequent category

Exercise 2

- Plot boxplot and histogram for Height and Score
- Overlay mean and median on histogram
- Interpret the distribution

Exercise 2 - Solution

```
library(ggplot2)

# Boxplot
ggplot(data, aes(y = Height)) +
  geom_boxplot(fill = "lightblue") +
  labs(title = "Boxplot of Height")

# Histogram with mean & median
ggplot(data, aes(x = Height)) +
  geom_histogram(binwidth = 5, fill = "lightgreen", color = "black") +
  geom_vline(aes(xintercept = mean(Height)), color = "red", linetype="dashed") +
  geom_vline(aes(xintercept = median(Height)), color = "blue", linetype="dotted") +
  labs(title = "Histogram of Height with Mean (red) and Median (blue)")
```

Interpretation:

- Boxplot shows median, IQR, whiskers, and outliers
- Histogram indicates skewness
- If mean > median → slight right skew

Exercise 3

- For a skewed variable in your dataset:
 - Compare mean and median
 - Compute SD and IQR
 - Explain which measure better represents “typical” value

Exercise 3 - Solution

```
# Suppose Score is skewed  
mean(data$Score)  
median(data$Score)  
sd(data$Score)  
IQR(data$Score)
```

Interpretation:

- Mean $>$ Median $\rightarrow$ right skewed
- SD = ... $\rightarrow$ spread of scores
- IQR = ... $\rightarrow$ central 50% range
- Median and IQR better represent typical values due to skewness